Problem 11.10

Suppose $\delta_t = \dfrac{t^3}{100}$. Find $\dfrac{1}{a(3)}$.

Solution

The force of interest and the accumulation function satisfy

$$\delta_t = \frac{a'(t)}{a(t)} = \frac{d}{dt} \ln a(t).$$

Since $a(0) = 1$, integrating from 0 to t gives

$$\ln a(t) = \int_0^t \delta_s \, ds.$$

Substitute the given force of interest:

$$\ln a(t) = \int_0^t \frac{s^3}{100} \, ds = \left. \frac{s^4}{400} \right|_0^t = \frac{t^4}{400}.$$

Therefore, the accumulation function is

$$a(t) = e^{t^4/400}.$$

At time 3,

$$a(3) = e^{3^4/400} = e^{81/400}.$$

Taking the reciprocal gives the discount factor:

$$\frac{1}{a(3)} = e^{-81/400} = e^{-0.2025}.$$

Hence,

$$\boxed{\frac{1}{a(3)} = e^{-81/400} \approx 0.81668648}.$$